

VSB ENGINEERING COLLEGE

Department of Computer Science & Academic Administration

TECHNICAL SPECIFICATION & TECHNOLOGY STACK REFERENCE MANUAL

1. Executive Project Summary

The VSB Academic Portal is an enterprise-grade Academic Performance Analytics and Automated Parent Communication Platform. It was engineered to solve three critical institutional challenges in higher education: (1) Eliminating manual, error-prone paper mark aggregation, (2) Providing tamper-proof NBA & NAAC-compliant grade audit trails, and (3) Delivering instant multi-channel progress alerts to parents via automated SMS and WhatsApp gateways.

2. High-Level Architectural Topology

The system is structured as a modern 3-Tier Enterprise Web Application designed for horizontal scalability, zero-trust role isolation, and sub-100ms API response times:

- **Tier 1: Single Page Application (SPA):** Responsive React 18 frontend built with TypeScript and Vite. Implements strict client-side role guards and dark-mode aesthetic.

- **Tier 2: API & Application Engine:** High-throughput Node.js & Express 5 RESTful micro-architecture with rate limiting, Zod schema validation, and JWT authentication.

- **Tier 3: Relational Persistence Tier:** Managed PostgreSQL 16 relational database interfaced through Prisma ORM with connection pooling, indexes, and ACID transactions.

- **Asynchronous Job Queue Tier:** Redis-backed BullMQ message queue for background SMS/WhatsApp dispatching without blocking web request cycles.

3. Frontend Technology Stack Breakdown

The client-side application is optimized for zero-latency interactions, cross-device responsiveness, and accessibility:

CLIENT FRAMEWORK	React & React DOM	v18.3.1
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Component-based reactive UI library utilizing hooks, concurrent rendering, and context-based state management.

LANGUAGE	TypeScript	v5.7.3
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Strongly-typed superset of JavaScript enforcing compile-time type safety across all components and API responses.

BUILD TOOL	Vite	v6.1.0
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Next-generation frontend tooling providing lightning-fast Hot Module Replacement (HMR) and optimized Rollup bundling.

CSS STYLING	Tailwind CSS & PostCSS	v3.4.17
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Utility-first CSS framework enabling high-density, custom dark-mode UI with zero runtime style overhead.

DATA VISUALIZATION	Recharts	v2.15.1
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Composable charting library built on SVG elements for rendering interactive department performance analytics.

ICONOGRAPHY	Lucide React	v0.475.0
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Clean, consistent vector icon set optimized for modern institutional dashboards and navigation menus.

4. Backend & Server Technology Stack

The server layer handles institutional security, grade calculations, PDF generation, and automated cloud deployments:

RUNTIME	Node.js	v20 LTS / 24
Asynchronous event-driven JavaScript runtime executing server-side logic with non-blocking I/O operations.		

WEB FRAMEWORK	Express.js	v5.2.1
Lightweight, flexible web framework powering the RESTful API endpoints and static client asset delivery.		

ORM / QUERY ENGINE	Prisma ORM & Prisma Client	v6.4.1
Next-generation type-safe Object-Relational Mapping library with declarative schema modeling and auto-migrations.		

AUTHENTICATION	JSON Web Tokens (jsonwebtoken)	v9.0.3
Stateless, signed bearer token standard implementing 24-hour expiration and cryptographic verification.		

PASSWORD HASHING	Bcrypt.js	v3.0.2
Adaptive cryptographic hash function utilizing 10 salt rounds to ensure zero plaintext credential storage.		

DATA VALIDATION	Zod	v4.6.1
TypeScript-first schema declaration and runtime validation library ensuring strict boundary sanitization.		

EXCEL / CSV PARSING	XLSX (SheetJS)	v0.18.5
High-speed spreadsheet parser extracting thousands of student roster rows from uploaded Excel and CSV workbooks.		

FILE UPLOADS	Multer	v2.3.0
Express middleware handling multipart/form-data with in-memory streaming buffers and size constraints.		

5. Document Generation & Presentation Engines

The platform features built-in native reporting engines that create executive-level exports on demand without third-party services:

PDF GENERATOR	PDFKit	v0.20.2
Server-side vector PDF document generation library for producing certified grade sheets, toppers rosters, and manuals.		

PRESENTATION DECKS	PptxGenJS	v4.0.1
Automated Microsoft PowerPoint (.pptx) generator creating slide decks for Board of Studies and HOD review meetings.		

6. Asynchronous Queue & Messaging Subsystem

JOB QUEUE	BullMQ	v6.3.4
Redis-based robust message queue managing delayed jobs, retry policies, and background notification dispatches.		

REDIS CLIENT	ioredis	v6.0.0
High-performance Redis client with built-in connection failover, lazy connecting, and graceful offline degradation.		

7. Relational Database Schema Design (PostgreSQL 16)

The PostgreSQL schema is structured to ensure complete relational integrity and support complex academic queries:

Table Name	Primary Key	Key Relational Fields	Purpose & Description
User	UUID	email, role, departmentId	Stores Admin & Faculty credentials and department mappings
Department	UUID	code (CSE, ECE, MECH)	Academic departments offering degree programs
AcademicYear	UUID	yearName (2025-2026), isCurrent	Academic operational sessions and active flags
Subject	UUID	code, semester, departmentId	Curriculum courses with maximum mark boundaries
Student	UUID	registerNumber, parentMobile	Enrolled student profiles and parent contact data
TeacherAssignment	UUID	staffId, subjectId, sectionId	Granular mapping unlocking grading access for faculty
Mark	UUID	studentId, subjectId, assessmentId	Recorded scores with unique constraint per student/exam
MarkChangeLog	UUID	markId, changedBy, reason	Tamper-evident audit trail recording previous/new marks
CollegeNotice	UUID	noticeType, targetType, channels	Institutional announcements and circular broadcasts
Notification	UUID	studentId, status, parentMobile	SMS & WhatsApp dispatch records and delivery statuses
AuditLog	UUID	userId, action, entityId	Chronological security log recording all administrative actions

8. Security Architecture & RBAC Matrix

The portal enforces an enterprise-grade defense-in-depth security model:

- **Role-Based Access Control (RBAC):** Enforces strict boundary separation between ADMIN (Institution-wide rights) and STAFF (restricted to assigned classes only).
- **Tamper-Proof Audit Logging:** Any grade modification after initial entry requires a mandatory justification. The system logs former marks, revised marks, operator identity, and timestamp.
- **Protection Against Auto-Fill Leakage:** The authentication views implement autoComplete="off" and autoComplete="new-password" to prevent browser password managers from auto-populating sensitive fields.
- **Security Headers & Rate Limiting:** HTTP security headers (HSTS, Content-Type protection) and IP-based rate limiting on /api endpoints safeguard against brute-force attacks.

9. Cloud DevOps, Containerization & Hosting Architecture

The application is engineered for zero-configuration, 1-click cloud deployment across leading modern platforms:

Containerization & Multi-Stage Dockerfile

The repository contains an enterprise 3-stage Dockerfile designed for minimal image size and non-root execution:

- **Stage 1 (client-builder):** Builds the React frontend using Node 20 Alpine and Vite, compiling minified assets into /client/dist.
- **Stage 2 (server-builder):** Generates the Prisma ORM query engine and compiles TypeScript server files into /server/dist.
- **Stage 3 (runner):** Combines static assets and server output, creates a dedicated unprivileged user (appuser:appgroup), and runs on Node 20 Alpine.

Railway Cloud Deployment (Active Live Production)

The live platform is deployed on Railway utilizing private container networking and managed PostgreSQL:

- **Public Live Web Service:** URL: <https://remarkable-exploration-production-ab2a.up.railway.app>
- **Internal Database:** PostgreSQL 16 running on postgres.railway.internal:5432 with persistent volume storage.
- **Automated Bootstrapping:** The container entrypoint executes "prisma db push" and auto-seeds initial master records on startup.

Render Cloud Deployment (Blueprint Ready)

The repository includes a certified render.yaml Blueprint enabling 1-click provisioning on Render:

- **Unified Root Build Script:** "npm run build" installs dependencies with --include=dev, builds client, generates Prisma, and compiles the server.
- **Health Probe:** Continuous HTTP health check on /health returning status 200 and database connectivity status.

10. Technology Summary Matrix

Layer	Primary Technologies	Key Advantage & Role
Frontend UI	React 18, Vite, TypeScript, Tailwind	Blazing-fast responsive SPA with type safety
Backend API	Node.js, Express 5, TypeScript, Zod	Robust RESTful API with boundary schema validation
Database Tier	PostgreSQL 16, Prisma ORM	ACID-compliant relational data modeling with zero SQL injection
Queues & Cache	BullMQ, Redis (ioredis)	Non-blocking asynchronous parent notification dispatching
Reporting	PDFKit, PptxGenJS, SheetJS (XLSX)	Native on-demand PDF, PPTX slides, and Excel cohort imports
Container & Cloud	Docker, Railway, Render Blueprint	Reproducible multi-stage container builds with 1-click CI/CD